



COVENANT UNIVERSITY
CANAANLAND, KM 10, IDIROKO ROAD
P.M.B 1023, OTA, OGUN STATE, NIGERIA.

TITLE OF EXAMINATION: B.Sc. Degree Examination

COLLEGE: College of Science and Technology

DEPARTMENT: Computer and Information Sciences

SESSION: 2021/2022

COURSE CODE: CSC213

COURSE TITLE: Structured Programming

INSTRUCTION: Answer any **FOUR(4)** Questions

SEMESTER: ALPHA

CREDIT UNIT: 3

TIME: 3 Hours

Question One

- a. Distinguish between a Pseudocode and Flowchart [2 Marks]
- b. Write a Python Program that outputs the following texts [3 Marks]
 - i. Python's the best!
 - ii. The cat said "meow."
 - iii. Read "Joshua and the Battle of Jericho" by tomorrow.
- c. Consider the Code below [3.5 Marks]
 - i. Identify and Correct the Errors in the following Code
 - ii. How many lines of names will be contained in the philosophers.txt when the code is executed?

```
1 outfile = openfile('philosophers.txt', 'r')
2 outfile.write('John Locke\n')
3 outfile.write('David Hume\n')
4 outfile.write('Edmund Burke\n')
5 for i in range(3)
6     name=input("Enter Philosper Name:\t")
7     outfile.write("name")
8 outfile.closefile()
```

- d. Many companies use telephone numbers like 555-GET-FOOD so the number is easier for their customers to remember. On a standard telephone, the alphabetic letters are mapped to numbers in the following fashion:

A, B, and C - 2

J, K, and L - 5

T, U, and V - 8

D, E, and F - 3

M, N, and O - 6

W, X, Y, and Z - 9

G, H, and I - 4

P, Q, R, and S - 7

Write a program that asks the user to enter a 10-character telephone number in the format XXX-XXX-XXXX. The application should display the telephone number with any alphabetic characters that appeared in the original translated to their numeric equivalent. For example, if the user enters 555-GET-FOOD, the application should display 555-438-3663.

[9 Marks]

Question Two

- a. A sequence is an object that holds multiple items of data, stored one after the other. Define and Mention the Characteristics of Two(2) types of sequence in Python. [2 Marks]
- b. Write a Python Programme that performs the following [3 Marks]
- Converts the List to a Tuple: `myList = ['Matthew', 'Mark', 'Luke', 'John']`
 - Assume the variable `name` references a string. Write a for loop that prints each character in the string
 - Assume the following statement appears in a program: `values = 'one$two$three$four'` Write a statement that splits the string, creating the following list: `['one', 'two', 'three', 'four']`
- c. Identify and Correct the Errors in the Following Code [3.5 Marks]

```
1 def main():
2     l_cities = 'Port-Harcourt Abuja Markurdi Kano'
3     cities= cities.split(" ")
4     outfile = open('cities.text', 'w')
5     for item in cities:
6         outfile.write(cities, '\n')
7     outfile.close()
8     __main__()
```

- d. Quadratic equation is made from a Latin term "quadrates" which means square. It is a special type of equation having the form of: ax^2+bx+c . Write a Python program that accepts the coefficient values of `a`, `b`, and `c` and solves the values of `x`. [9 Marks]

Question Three

- a. Define the following Python Concepts with examples [3 Marks]
- Sequence Structures
 - Control Structures
 - Repetition Structure
- b. What is the output of the following Code? [3 Marks]

```
i) animal = 'Lion'
   animal[0] = 'Z'
   print (animal)

ii) letters = 'ABCDEFGHIJKLMNOPQRSTUVWXYZ'
   print(letters[0:26:2])

iii) stuff = {1 : 3, 2 : 2, 3 : 1}
   print(stuff[3])
```

- c. Identify and Correct the Errors in the Following Code [2.5Marks]

```
1 import logn
2 def main():
3     password = input('Enter your password: ')
4     while not login.valid_password(password)
5         print('That password is not valid.')
6     pasword = input('Enter your password: ')
7     print('That is a valid password.')
8     __main__()
```

- d. Assume a file containing a series of integers is named numbers.txt and exists on the computer's disk. Write a program that reads all of the numbers stored in the file and calculates their Maximum, Minimum, Total, and Average. [9 Marks]

Question Four

- a. Explain the following Python Concepts with relevant Examples

- i. Slicing
- ii. Global Variable

[2 Marks]

- b. [3 Marks] What is the Output of the Following Code

```
i) L = [10, 20, 30, 40]
   print(L[-1], L[-2], L[-3], L[-4]) ii) for num in list(range(1, 10, 2)):
                                       print(num)

   for number in range(1, 5):
iii)     square = math.sqrt(number)
         print(number, '\t', square)
```

- c. Describe Three (3) Dictionary Functions with examples

[3 Marks]

- d. Students are assigned a portal login ID to log into and use the University portal. As part of your internship with the university's CSIS, you have been asked to write the code that generates portal login ID for students. You will use the following algorithm to generate a login ID:

- i. Get the first three characters of the student's first name. (If the first name is less than three characters in length, use the entire first name.)
- ii. Get the first three characters of the student's last name. (If the last name is less than three characters in length, use the entire last name.)
- iii. Get the last three characters of the student's Matric Number. (If the Matric number is less than three characters in length, use the entire Matric number.)
- iv. Concatenate the three sets of characters to generate the portal login ID.

For example, if a student's name is Ayanfe Adebisi, and her Matric number is 20CA023345, her login ID would be AyaAde345.

[9.5 Marks]

Question Five

- a. Define the following concepts in Python

- i. Functions
- ii. Dictionary

[2 Marks]

- b. [3 Marks] What is the Output of the Following Code

```
i) myset = set('aaabbbccc')
   print(myset) ii) newset = set('one two three')
                  print(newset)

iii) print(format(987654.129, ',.5f'))
```

c. [3.5 Marks] Identify and Correct the Errors in the Following Code

```
1 starlist = ()
2 again == 'y'
3 while again.lower() = 'y':
4     fname = input('Enter your Favorite Bible Character: ')
5     star_list.append(name)
6     print('Do you want to add another name?')
7     again == input('y = yes, anything else = no: ')
8 print()
9 print('Here are the names you entered.')
10 for name in star_list
11     print(name)
```

d. Given the following Sets `choir_members = set(['Ade', 'Amina', 'Heida', 'Segun'])`; `TC_members = set(['Amina', 'Onyeka', 'Bola', 'Ade'])`, write a Python programme that uses Python's Set Operators to perform the following operations:

- i. Display members of the Choir.
- ii. Display Members of TC
- iii. Display members that belong to Choir and TC
- iv. Display members of either Choir or TC
- v. Display members of Choir but not TC
- vi. Display members of TC but not Choir
- vii. Display members of Service Unit but not both

[9 Marks]

Question Six

- a. Distinguish between an **Argument** and **Parameter** in Python's function [1.5 Marks]
- b. Describe the **File Modes** in Python [1.5 Marks]
- c. Write a program that opens an output file with the filename things.txt, writes the name of an animal, a fruit, and a country to the file on separate lines, then closes the file [3.5 Marks]
- d. Identify and Correct the Errors in the Following Code [3 Marks]

```
~1 prod_nums = [V475, F987, Q143, R688]
2 search = input('Enter a product number:')
3 if search == prod_nums:
4     print('search was found in the list.')
5 else
~6     print(search, 'was not found in the list'.)
```

- e. Dr. Shuther teaches a Programming class and uses the following 5-point grading scale for all his exams: He has asked you to write a program that will allow a student to enter a test score and then display the grade for that score; such that 90 and above is A; 80 - 90 is B; 70 - 79 is C; 60 - 69 is D; Below 60 is F. [8 Marks]